

Davide Revignas

Curriculum Vitae et Studiorum

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Personal Data

Born Palmanova (Italy), 13 March 1995
Languages Italian, Friulian (native speaker), English (fluent), French (intermediate)

Research Experience

- Dec. 2024 – **Postdoctoral Researcher**, *Biological, Soft Matter and Statistical Physics Group*,
present Department of Physics and Astronomy, University of Padua, Italy.
Project: *Unraveling the folding mechanisms of entangled proteins.*
Supervisor: Prof. A. Trovato.
- Oct. 2023 – **Visiting Scientist**, *Biological Physics Group*, Max Planck Institute for the
Oct. 2024 Physics of Complex Systems, Dresden, Germany.
Project: *The role of water in the spatiotemporal organization of biomolecules in cells.*
Supervisors: Prof. G. Franzese and Prof. F. Jülicher.
- Jan. 2023 – **Research Fellow**, *Soft Matter Theory Group*, Department of Chemical Sciences,
Sep. 2023 University of Padua, Italy.
Project: *Numerical simulation of the mechanical properties of polymeric systems.*
Supervisor: Prof. A. Ferrarini.

Education

- Jun. 2023 **Ph.D. in Science and Engineering of Materials and Nanostructures**,
University of Padua, Padua, Italy, Final grade: excellent *cum laude*.
Thesis: *Curvature, Flexibility and Orientational Fluctuations: How Molecular Features Translate into Elastic Properties and Mesoscale Structure in Nematics.*
Supervisor: Prof. Alberta Ferrarini.
- Sept. 2019 **M.Sc. in Materials Science**, University of Padua, Padua, Italy, Final grade:
110/110 *cum laude*.
Thesis: *Elasticity of Polymeric Liquid Crystals.*
Supervisor: Prof. Alberta Ferrarini.
- Aug. 2018 – **Erasmus+ Exchange**, Aarhus University, Aarhus, Denmark.
Feb. 2019 Coursework in the Departments of Physics, Chemistry and in the Interdisciplinary Nanoscience Center.
- Oct. 2017 **B.Sc. in Materials Science**, University of Padua, Padua, Italy, Final grade:
110/110 *cum laude*.
Project: *Diamond Growth by CVD.*
Supervisor: Prof. Giovanni Mattei.

Publications

- 2025 G. Bartolucci, D. M. Busiello, M. Ciarchi, A. Corticelli, I. Di Terlizzi, F. Olmeda,
D. Revignas and V. M. Schimmenti, *Phase behavior of Cacio e Pepe sauce.*
Physics of Fluids **37**, 044122 (2025).

- 2024 S. Varytimiadou, **D. Revignas**, F. Giesselmann and A. Ferrarini, *Elasticity of lyotropic nematic liquid crystals: A review of experiments, theory and simulation. Liquid Crystals Reviews* **12**, 57–104 (2024).
- 2024 **D. Revignas** and A. Ferrarini, *Molecular shape, elastic constants and spontaneous twist in chiral and achiral nematics: Insights from a generalised Maier–Saupe framework. Liquid Crystals* **51**, 898–918 (2024).
- 2023 **D. Revignas** and A. Ferrarini, *On the elusive saddle–splay and splay–bend elastic constants of nematic liquid crystals. The Journal of Chemical Physics* **159**, 034905 (2023).
- 2023 **D. Revignas** and A. Ferrarini, *Spontaneous twisting of achiral hard rod nematics. Physical Review Letters* **130**, 028102 (2023).
- 2022 **D. Revignas** and A. Ferrarini, *Microscopic modelling of nematic elastic constants beyond Straley theory. Soft Matter* **18**, 648–661 (2022).
- 2022 **D. Revignas** and V. Amendola, *Artificial neural networks applied to colorimetric nanosensors: An undergraduate experience tailorable from gold nanoparticles synthesis to optical spectroscopy and machine learning. Journal of Chemical Education* **99**, 2112–2120 (2022).
- 2021 **D. Revignas** and A. Ferrarini, *From bend to splay dominated elasticity in nematics. Crystals* **11**, 831 (2021).
- 2020 **D. Revignas** and A. Ferrarini, *Interplay of particle morphology and director distortions in nematic fluids. Physical Review Letters* **125**, 267802 (2020).

--- Honors and Scholarships

- 2025 Ig Nobel Prize in Physics, shared with the seven co-authors of the paper *Phase Behavior of Cacio e Pepe Sauce*.
- 2024 Commendation for the quality of the Ph.D. thesis by the SICL “Daniela Pucci” Award Committee.
- 2019–2023 CARIPARO Ph.D. Scholarship, awarded as best Ph.D. applicant in the 35th cycle of the Ph.D. programme in Science and Engineering of Materials and Nanostructures.
- 2014 EDIPOWER Merit Scholarship

--- Presentations

- 2026 **D. Revignas** and A. Ferrarini, *Microscopic modeling of nematic elastic constants*. Invited presentation at the COMPUTATIONAL SOFT MATTER LAB SEMINARS. Amsterdam, The Netherlands, 16 June 2026.
- 2026 **D. Revignas** and A. Trovato, *Entangled motifs impose specific constraints on good-folding sequences in lattice protein models*. Oral presentation at the 9TH WORKSHOP PHYSICS OF BIOMOLECULES: STRUCTURE, DYNAMICS AND FUNCTION. Bressanone, Italy, 11 February 2026.
- 2025 **D. Revignas**, C. De Michele and A. Ferrarini, *Generalized Onsager theory for nematic polymers*. Oral presentation in the session *Novel Views on Statistical Physics of Polymer Conformations and Dynamics* at FISMAT 2025. Venice, Italy, 11 July 2025.
- 2024 **D. Revignas** and A. Ferrarini, *Microscopic insights on the puzzling saddle–splay elasticity of nematics*. Oral presentation at the SPRING MEETING OF THE GERMAN PHYSICAL SOCIETY. Berlin, Germany, 21 March 2024.
- 2022 **D. Revignas** and A. Ferrarini, *The elusive saddle–splay elasticity of nematics*. Oral presentation at the 28TH INTERNATIONAL LIQUID CRYSTAL CONFERENCE. Lisbon, Portugal, 26 July 2022.

- 2022 **D. Revignas** and A. Ferrarini, *The puzzling saddle-splay elasticity of nematics: Do rods spontaneously twist?* Oral presentation at ITALIAN SOFT DAYS 2022. Bari, Italy, 29 September 2022.
- 2021 **D. Revignas** and A. Ferrarini, *How shape polarity and concavity of mesogens affect Frank elastic constants.* Oral presentation at the 18TH INTERNATIONAL CONFERENCE ON FERROELECTRIC LIQUID CRYSTALS. Ljubljana, Slovenia, 9 September 2021.

Posters

- 2025 **D. Revignas** and A. Trovato, *Do entangled lasso motifs affect protein design? Lessons from a lattice protein model.* Poster presented at the SUMMER SCHOOL ON MOLECULAR DESIGN: FROM COMPUTATIONAL MODELS TO BIOLOGICAL AND MEDICAL APPLICATIONS. Cargèse, France, 18–29 August 2025.
- 2025 **D. Revignas** and A. Ferrarini, *A generalised Maier–Saupe framework for twisted nematics.* Poster presented at the 29TH INTERNATIONAL CONFERENCE ON STATISTICAL PHYSICS (STATPHYS 29). Florence, Italy, 13–18 July 2025.
- 2023 **D. Revignas** and A. Ferrarini, *Spontaneous twisting of achiral nematic rods.* Poster presented at the TOPOLOGICAL SOFT MATTER WORKSHOP 2023. Edinburgh, United Kingdom, 10–12 May 2023.
- 2022 **D. Revignas** and A. Ferrarini, *Spontaneous twisting of achiral nematic rods.* Poster presented at the EUTOPIA SUMMER SCHOOL ON TOPOLOGY IN SOFT MATTER AND BIOLOGICAL MOLECULES. Paris, France, 27 June–6 July 2022.
- 2022 **D. Revignas** and A. Ferrarini, *Spontaneous twisting of achiral nematic rods.* Poster presented at the 14TH ITALIAN LIQUID CRYSTALS SOCIETY (SICL) CONFERENCE. Padua, Italy, 22–23 September 2022.
- 2021 **D. Revignas** and A. Ferrarini, *cDFT calculation of Frank constants for non-convex particles.* Poster presented at the 11TH LIQUID MATTER CONFERENCE. Online, 18–23 June 2021.
- 2020 **D. Revignas** and A. Ferrarini, *Elasticity of liquid crystals: Beyond Straley model.* Poster presented at ITALIAN SOFT DAYS 2020. Online, 21–25 September 2020.

Teaching and Supervision

- 2026 Co-supervision of the B.Sc. thesis in Physics *Folding of Proteins Characterized by Topological Complexity in Lattice Models*. Main supervisor: Prof. A. Trovato. Defended on 13 July 2026.
- 2025 Co-supervision of the M.Sc. thesis in Physics *Phase Behavior and Cluster Size Distribution in Polymer Solutions: From Homopolymers to Heteropolymers*. Main supervisor: Prof. A. Maritan. Defended on 20 October 2025.
- 2022 Teaching assistant for the *Physical Chemistry Laboratory* course, B.Sc. in Industrial Chemistry, University of Padua, 20 hours.
- 2021 Lecturer at the multidisciplinary workshop *Colors of Nature* for high school students, approximately 4 hours of lectures and laboratory activities.
- 2021 Teaching assistant for the *Physical Chemistry Laboratory* course, B.Sc. in Industrial Chemistry, University of Padua, 20 hours.

Public Outreach and Organizing Activities

- 2026 **D. Revignas** and D. M. Busiello, *La Cacio e Pepe da Ig Nobel Prize.* Presentation and cooking demonstration at the 10th Food and Science Festival. Mantua, Italy, 16 May 2026.

- 2025 **D. Revignas**, “Una Cacio e Pepe da Ig Nobel”, article published online in *SIF Prima Pagina*, the online magazine of the Italian Physical Society, 28 November 2025.
- 2025 **D. Revignas** and D. M. Busiello, *Show Cooking Cacio e Pepe*. Presentation and cooking demonstration at the Science4All Festival. Padua, Italy, 27 September 2025.
- 2022 Member of the organizing committee of the 14TH ITALIAN LIQUID CRYSTALS SOCIETY (SICL) CONFERENCE. Padua, Italy.

Refereeing Activity

Journals Physical Review Letters; Physical Review Research; The Journal of Chemical Physics; Physical Review E; Journal of Statistical Mechanics: Theory and Experiment.

Co-authored Presentations

* *The underlined name indicates the presenter.*

- 2026 **D. Revignas** and A. Trovato, *Entangled motifs generate energetic frustration in good-folding sequences within lattice protein models*. Oral presentation at 12TH INTERNATIONAL CONFERENCE ON BIOLOGICAL PHYSICS. Genoa, Italy, 2 July 2026.
- 2026 **D. Revignas** and A. Trovato, *Entangled motifs generate energetic frustration in good-folding sequences within lattice protein models*. Oral presentation at BIOLOGICAL PHYSICS AND STATISTICAL MECHANICS. Turin, Italy, 9 June 2026.
- 2023 **D. Revignas** and A. Ferrarini, *Some open questions in the elasticity of nematic liquid crystals: Insights from microscopic theory*. Invited presentation at the 36TH MARIAN SMOLUCHOWSKI SYMPOSIUM ON STATISTICAL PHYSICS. Krakow, Poland, 27 September 2023.
- 2023 **D. Revignas** and A. Ferrarini, *Saddle-Splay Elasticity and Chiral Structures in Confined Nematics*. Oral presentation at the 49TH GERMAN LIQUID CRYSTAL CONFERENCE. Stuttgart, Germany, 24 March 2023.
- 2022 **D. Revignas** and A. Ferrarini, *Twisted ground state in achiral anisotropic fluids*. Invited presentation at the COMPLEX PHASES IN SOFT MATTER SYMPOSIUM, organized by the Condensed Matter Division of the EPS. Manchester, United Kingdom, 21–26 August 2022.
- 2022 **D. Revignas** and A. Ferrarini, *Shape polarity, chirality and curvature as players in the behavior of nematics*. Invited presentation at the 28TH INTERNATIONAL LIQUID CRYSTAL CONFERENCE. Lisbon, Portugal, 25 July 2022.
- 2021 **D. Revignas**, C. Greco and A. Ferrarini, *Molecular features and polar nematic order*. Invited presentation at the 18TH INTERNATIONAL CONFERENCE ON FERROELECTRIC LIQUID CRYSTALS. Ljubljana, Slovenia, 9 September 2021.
- 2021 **D. Revignas** and A. Ferrarini, *Elasticity of liquid crystals: Beyond Straley theory*. Invited presentation at the 11TH LIQUID MATTER CONFERENCE. Online, 18–23 June 2021.

Attended Workshops and Schools

- 2025 Molecular Design: From Computational Models to Biological and Medical Applications. Cargèse, France, 18–29 August 2025.
- 2023 Topological Soft Matter 2023. Edinburgh, United Kingdom, 10–12 May 2023.

- 2022 EUTOPIA Summer School on Topology in Soft Matter and Biological Molecules. Paris, France, 27 June–6 July 2022.
- 2021 IOP Advanced School in Soft Condensed Matter. Online, 5–9 July 2021.
- 2021 7th UMass Summer School on Soft Solids and Complex Fluids. Online, 1–4 June 2021.
- 2020 Frontiers in Computational Methods for Active Matter. Lausanne, Switzerland, 10–12 February 2020.
- 2019 Optical Tweezers: Biophysics and Beyond. Padua, Italy, 25–26 November 2019.

Computational Skills

Coding Python, Fortran, Bash, and MATLAB (advanced); C++, Mathematica (basic).
Software LAMMPS, Quantum ESPRESSO, VMD, Git, GitHub, and $\text{\LaTeX}$; Slurm and HTCondor for high-performance computing job management.